Dynamic Model — Dialog Map

SC2006 Team 4 · Dialysis Transport Coordination System · Lab 2 Deliverable

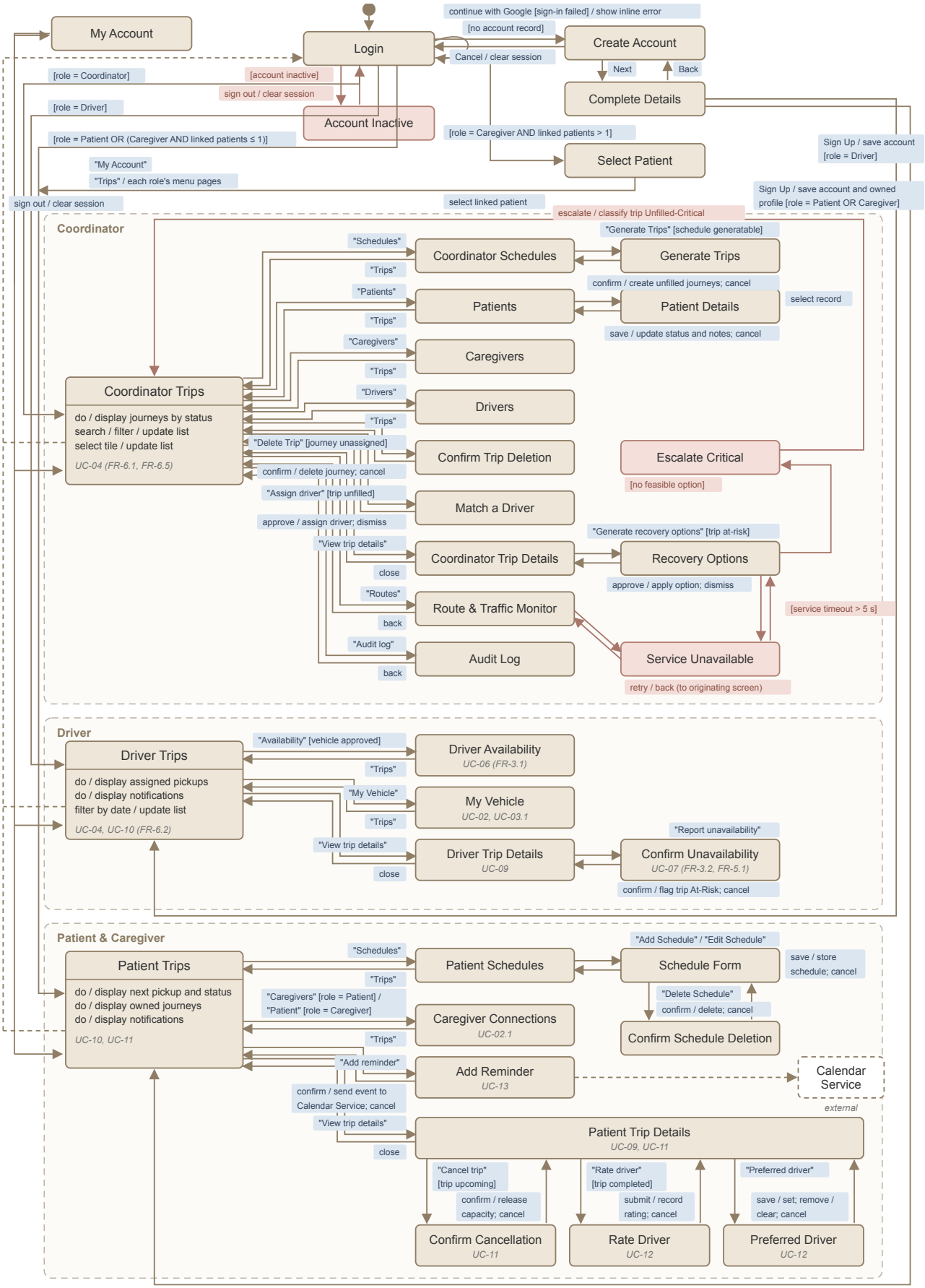
This document models the system's behaviour as a state machine of user interface screens, following Lab 2 §3.3.1. States are UI screens; transitions are user actions or system events. The model follows the routes and screens the system provides; states that are designed but have no data path behind them yet are identified in the design notes.

Modelling conventions

- A screen is a separate **state** only where the set of available actions changes. Filtering, sorting and expanding content in place are **internal transitions**, written inside the state box as `event / action`.
- `do /` denotes a continuous activity while the state is active. Event-driven behaviour that completes is written as an internal transition, not as `do /`.
- Guards are written in square brackets on the transition to which they apply.
- Exception and failure states are shaded differently from normal UI states.
- External systems outside the system boundary are drawn with a dashed outline and are not states.

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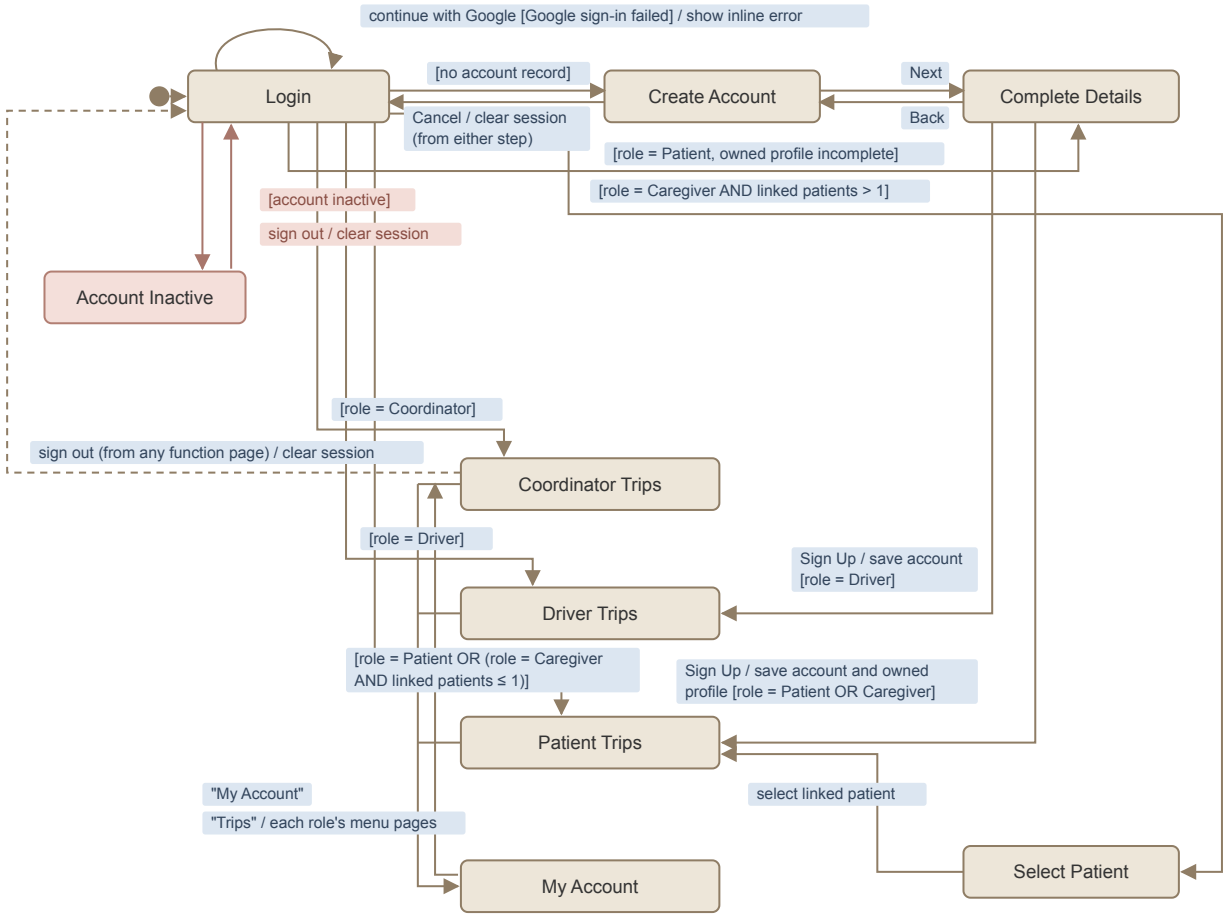
1. Complete Dialog Map



How to read this map

- This is the whole model in one diagram. Login, Create Account, Complete Details, Account Inactive, Select Patient and My Account belong to no role map and are shared by every role; the three dashed groupings are the Coordinator, Driver and Patient/Caregiver maps reached after authentication. The groupings are drawing conventions, not states.
- Sections 3–5 redraw each grouping on its own, at a larger size and with its own design notes. No state or transition appears in those sections that is not in this map, and none in this map is omitted from them. Login and My Account appear there as the shared targets they are; both are defined in Section 2.
- The guarded transitions out of Login determine which map the session enters. Each authenticated session is bound to a single role (FR-1.3), so only one grouping is reachable per session.
- Sign-out is drawn once, as the dashed line running from all three entry states back to Login, rather than repeated on every state.
- Section 6 lists every transition in tabular form and is the formal statement of the model.

2. Shared Entry



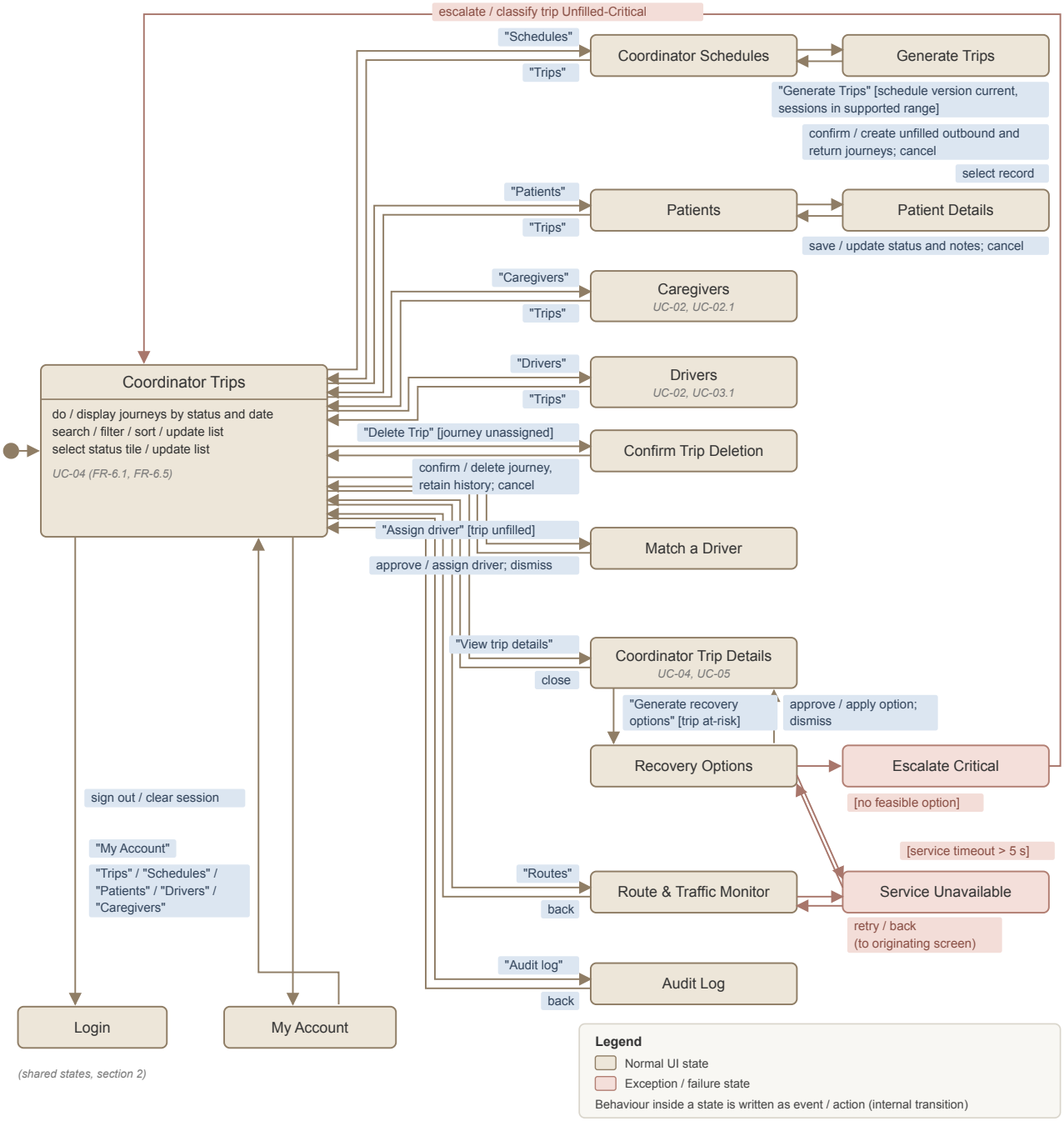
Dashed arrow: sign out from any function page of any role map returns here.
Login, Create Account, Complete Details, Account Inactive, Select Patient and My Account belong to no role map; every active role reaches My Account.
Coordinator accounts are provisioned manually, so no sign-up path enters the Coordinator map.

Design notes

- Coordinator Trips, Driver Trips and Patient Trips are the entry points into the three role maps in Sections 3–5. They are one route, `/dashboard/trips`, rendered for the role of the session; `/dashboard` only redirects to it, so the model carries no separate landing state.

- Authentication is Google OAuth, so Login carries no credential form. A failed sign-in is modelled as a **self-transition** on Login carrying the action `show inline error`. While the service health check fails, the sign-in control is disabled; that is a condition of the state, not a transition.
- **Create Account** and **Complete Details** are the two sign-up steps on the landing card. Next and Back only move between them and change no stored data; the account and, for a Patient, the owned profile are saved atomically on Sign Up. Only Patient, Caregiver and Driver can be chosen, so no sign-up path enters the Coordinator map — coordinators are provisioned manually. A Patient whose owned profile is incomplete enters Complete Details instead, whether that is detected at Login or later on My Account.
- **Account Inactive** is a state rather than an inline message. An authenticated account that is not active reaches a screen whose only exit is sign out, so an inactive account cannot proceed into a role map.
- **My Account** belongs to no role map: every active role reaches it from any function page and returns through the same navigation. A Patient edits their owned details in place on this screen, so edit, save and cancel are internal transitions, not a separate form state. Notifications are likewise a header panel available on every function page, not a state.
- Caregivers authenticate through the same Patient entry point using their own account. A Caregiver linked to at most one Patient transitions directly to Patient Trips, under the same guarded transition as a Patient; a Caregiver linked to more than one Patient passes through **Select Patient** first, since the system cannot determine the scope of the session without that choice. Caregiver linking (UC-02.1) is implemented, but the shipped screens show a Caregiver every linked Patient's journeys together and choose the Patient inside the schedule form, so no screen yet performs this selection.
- Signing out from any function page returns to Login rather than to a final state, because the application does not terminate on sign-out. Each authenticated session is bound to a single role (FR-1.3), so only one role map is reachable per session.
- Related use case: UC-01 Authenticate User (FR-1.2, FR-1.3, NFR-5.1, NFR-5.2).

3. Coordinator

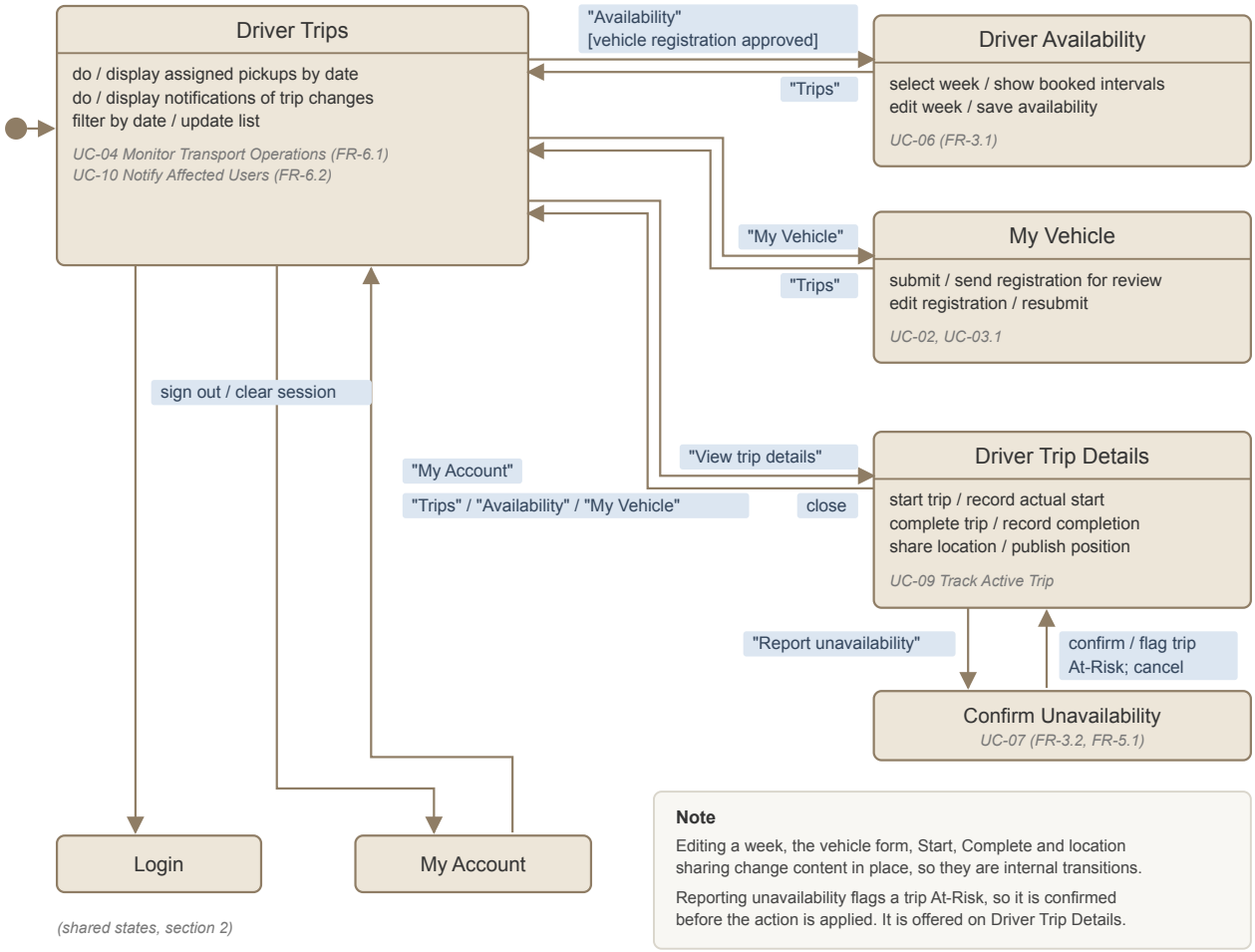


Design notes

- Coordinator Trips carries the UC-04 status categories. Its status tiles, search, filters, sorting and pagination change content within the same screen, so they are internal transitions, not states. The same holds for the Patients, Caregivers, Drivers and Coordinator Schedules lists, whose record-detail and edit dialogs also act in place.
- **Coordinator Trip Details** is the routed page `/dashboard/trips/:tripId`. It is a state because its actions differ from the list's, and it is drawn once per role map, as Trips is, because those actions differ by role. Recovery is offered only here, so Recovery Options is entered from it and returns to it.
- Guards are shown on the transitions: "Assign driver" is available only on Unfilled trips, "Generate recovery options" only on At-Risk trips, and "Delete Trip" only on a journey with no driver assigned.

- **Coordinator Schedules** is read-only: treatment requirements are recorded by the Patient, not the coordinator. **Generate Trips** replaces the New Trip state of the earlier model. Journeys are derived from a selected treatment schedule over a bounded future range using the treatment times and the standard 15-minute buffer (UC-03), rather than typed in by hand, and generation is idempotent.
- **Patients** covers UC-02. Record creation belongs to sign-up and deletion is not permitted, so Patient Details carries neither an "add new" transition nor an archive action; a coordinator may save only Status and Notes.
- **Confirm Trip Deletion** is a state for the same reason as the other confirmations: removing a journey cannot be undone from the list. It applies only to an unassigned journey, and the journey's history is retained.
- **Escalate Critical** covers FR-5.4. Where the system can generate no feasible recovery option, the trip is classified Unfilled-Critical and escalated for manual intervention rather than silently returning the coordinator to Coordinator Trips. The only exit is `escalate`, so a trip in this state cannot be left without being escalated.
- **Service Unavailable** covers the external-dependency failure implied by FR-4.1, FR-4.2 and NFR-2.1. Where routing or traffic data cannot be retrieved within the five-second bound, the coordinator is shown a recoverable failure state rather than stale or absent data. The same exception applies to Recovery Options, which depends on the same services; retry or back returns the coordinator to whichever of the two screens raised it.
- Match a Driver and Recovery Options are implemented. Route & Traffic Monitor and Audit Log are modelled as the use cases require them; the shipped app presents route, traffic and history data for each trip on its details page rather than as separate screens.
- **Caregivers** and **Drivers** are coordinator function pages reached through the header navigation. Driver account status, vehicle approval and rejection, and caregiver status are edited in place on them.
- The Audit Log serves UC-04 (alternative flow: view audit log) and FR-6.4.
- Related use cases: UC-02 Manage Patient Records, UC-03 Manage Treatment and Transport Schedules, UC-04 Monitor Transport Operations, UC-05 Resolve At-Risk Trip, UC-08 Assess Route and ETA.

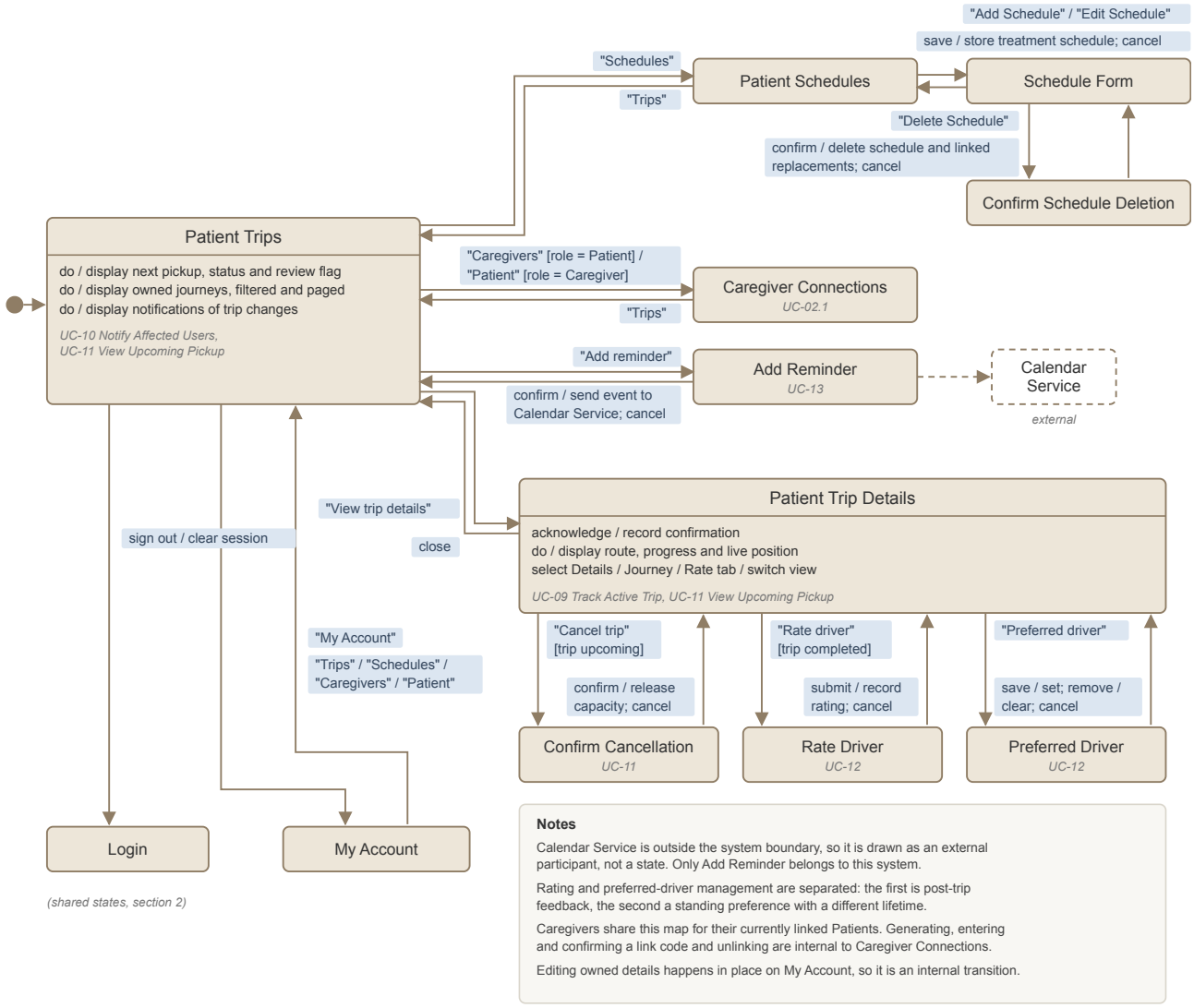
4. Driver



Design notes

- Driver Trips is the Trips function page rendered for a Driver: assigned pickups by date. When no pickups are assigned it says so rather than inventing trips.
- **Driver Availability** and **My Vehicle** are separate routes. Availability is reachable once a vehicle registration has been approved; before that the menu offers Register Vehicle, which is the same My Vehicle state. Editing a week is an internal transition (`edit week / save availability`), not a `do /` activity, because it is triggered by a user event and completes.
- **Driver Trip Details** carries Start, Complete and location sharing as internal transitions, and is where unavailability is reported.
- Notifications appear in-page, in the shared header panel, so they are modelled as a display activity on this state rather than as a separate screen.
- **Confirm Unavailability** is modelled as a state. Reporting unavailability flags the trip At-Risk and requires coordinator intervention (FR-5.1), so it is confirmed before the action is applied rather than being a single irreversible tap.
- Related use cases: UC-06 Manage Driver Availability, UC-07 Report Driver Unavailability, UC-10 Notify Affected Users.

5. Patient & Caregiver



Design notes

- Patient Trips is a single scrolling page carrying the next pickup — its pickup status, journey date and direction and any review flag — above a date-filtered, paged list of owned journeys. Route addresses, the treatment-derived constraints and live progress are on **Patient Trip Details**, the routed page `/dashboard/trips/:tripId`. There is no separate tracking screen.
- **Patient Schedules** serves both roles: a Caregiver reaches it for their linked Patients. **Caregiver Connections** is the linking page (UC-02.1); generating, entering and confirming an invitation code and unlinking are internal transitions on it. **Schedule Form** covers both adding and editing a treatment schedule; the whole-series / one-session scope choice inside it changes the same form, so it is an internal transition. **Confirm Schedule Deletion** is a state because deleting a series also deletes its linked replacements.
- Acknowledging an upcoming trip acts in place on Patient Trip Details and is an internal transition. Cancelling does not: **Confirm Cancellation** is a state, because cancelling a dialysis journey releases driver capacity and cannot be undone by the patient. It carries the guard `[trip upcoming]`, since a completed trip cannot be cancelled.
- **Rate Driver** carries the guard `[trip completed AND role = Patient]`. A trip that has not taken place cannot be rated, and rating is the Patient's own: a Caregiver is not routed to it. **Preferred**

Driver carries `[role = Patient]` for the same reason.

- Rating and preferred-driver management are modelled as two states rather than one. They have different triggers and different lifetimes: one is feedback on a completed trip, the other a standing preference that persists across trips. Both trace to UC-12.
- **Calendar Service** is outside the system boundary and is therefore drawn as an external participant, not as a state. The state belonging to this system is **Add Reminder**; the call to the external service is an action on the confirm transition. Calendar synchronisation is not implemented, so Add Reminder is designed but has no data path behind it yet.
- This map is shared by Patients and Caregivers under the login assumption in Section 2. A Caregiver acts only for currently linked Patients and may acknowledge or cancel their trips; rating and preferred-driver management stay the Patient's own.
- Related use cases: UC-09 Track Active Trip, UC-10 Notify Affected Users, UC-11 View Upcoming Pickup, UC-12 Manage Driver Feedback and Preference, UC-13 Manage Calendar Reminders.

6. Transition Summary

The transitions below are the formal statement of the model; the diagrams are their visual form. A dash in the Event column denotes a transition triggered by its guard becoming true rather than by a user action. Header navigation is listed once per role: "any Coordinator function page" is Coordinator Trips, Coordinator Schedules, Patients, Caregivers, Drivers or Coordinator Trip Details; "any Driver function page" is Driver Trips, Driver Availability, My Vehicle or Driver Trip Details; "any Patient or Caregiver function page" is Patient Trips, Patient Schedules, Caregiver Connections or Patient Trip Details.

From	Event	Guard	Action	To
Login	continue with Google	[account active, role = Coordinator]	—	Coordinator Trips
Login	continue with Google	[account active, role = Driver]	—	Driver Trips
Login	continue with Google	[account active, role = Patient, owned profile complete]	—	Patient Trips
Login	continue with Google	[role = Caregiver, linked patients ≤ 1]	—	Patient Trips
Login	continue with Google	[role = Caregiver, linked patients > 1]	—	Select Patient
Login	continue with Google	[no account record]	—	Create Account
Login	continue with Google	[role = Patient, owned profile incomplete]	—	Complete Details
Login	continue with Google	[account inactive]	—	Account Inactive
Login	continue with Google	[Google sign-in failed]	show inline error	Login (self)
Create Account	Next	—	—	Complete Details
Create Account	change Google account	—	restart Google selection	Create Account (self)
Create Account	Cancel	—	clear session	Login
Complete Details	Back	—	—	Create Account
Complete Details	Sign Up	[role = Patient]	save account and owned patient profile	Patient Trips
Complete Details	Sign Up	[role = Caregiver]	save account	Patient Trips
Complete Details	Sign Up	[role = Driver]	save account	Driver Trips
Complete Details	Cancel	—	clear session	Login
Account Inactive	sign out	—	clear session	Login
Select Patient	select linked patient	—	set session scope to patient	Patient Trips

From	Event	Guard	Action	To
any function page (all role maps)	"My Account"	—	—	My Account
My Account	"Trips"	[role = Coordinator]	—	Coordinator Trips
My Account	"Trips"	[role = Driver]	—	Driver Trips
My Account	"Trips"	[role = Patient OR Caregiver]	—	Patient Trips
My Account	"Schedules"	[role = Coordinator]	—	Coordinator Schedules
My Account	"Schedules"	[role = Patient OR Caregiver]	—	Patient Schedules
My Account	"Patients"	[role = Coordinator]	—	Patients
My Account	"Caregivers"	[role = Coordinator]	—	Caregivers
My Account	"Drivers"	[role = Coordinator]	—	Drivers
My Account	"Availability"	[role = Driver, vehicle registration approved]	—	Driver Availability
My Account	"My Vehicle"	[role = Driver]	—	My Vehicle
My Account	"Caregivers"	[role = Patient]	—	Caregiver Connections
My Account	"Patient"	[role = Caregiver]	—	Caregiver Connections
My Account	edit	[role = Patient]	unlock owned details	My Account (self)
My Account	save	[role = Patient, version current]	update owned profile	My Account (self)
My Account	cancel	[role = Patient]	discard changes	My Account (self)
My Account	—	[role = Patient, owned profile missing]	—	Complete Details
any Coordinator function page	"Trips"	—	—	Coordinator Trips
any Coordinator function page	"Schedules"	—	—	Coordinator Schedules
any Coordinator function page	"Patients"	—	—	Patients
any Coordinator function page	"Caregivers"	—	—	Caregivers
any Coordinator function page	"Drivers"	—	—	Drivers
Coordinator Schedules	"Generate Trips"	[schedule version current, sessions in supported range]	—	Generate Trips
Generate Trips	confirm	[future range ≤ 90 days]	create unfilled outbound and return journeys	Coordinator Schedules

From	Event	Guard	Action	To
Generate Trips	cancel	—	—	Coordinator Schedules
Coordinator Trips	"Delete Trip"	[journey unassigned]	—	Confirm Trip Deletion
Confirm Trip Deletion	confirm	[version current]	delete journey, retain history	Coordinator Trips
Confirm Trip Deletion	cancel	—	—	Coordinator Trips
Coordinator Trips	"Assign driver"	[trip unfilled]	—	Match a Driver
Match a Driver	approve	—	assign driver	Coordinator Trips
Match a Driver	dismiss	—	—	Coordinator Trips
Coordinator Trips	"View trip details"	—	—	Coordinator Trip Details
Coordinator Trip Details	close	—	—	Coordinator Trips
Coordinator Trip Details	"Generate recovery options"	[trip at-risk]	—	Recovery Options
Recovery Options	approve	—	apply recovery option	Coordinator Trip Details
Recovery Options	dismiss	—	—	Coordinator Trip Details
Recovery Options	—	[no feasible option]	—	Escalate Critical
Recovery Options	—	[service timeout > 5 s]	—	Service Unavailable
Escalate Critical	escalate	—	classify trip Unfilled-Critical	Coordinator Trips
Coordinator Trips	"Routes"	—	—	Route & Traffic Monitor
Route & Traffic Monitor	back	—	—	Coordinator Trips
Route & Traffic Monitor	—	[service timeout > 5 s]	—	Service Unavailable
Service Unavailable	retry / back	[entered from Route & Traffic Monitor]	—	Route & Traffic Monitor
Service Unavailable	retry / back	[entered from Recovery Options]	—	Recovery Options
Coordinator Trips	"Audit log"	—	—	Audit Log
Audit Log	back	—	—	Coordinator Trips
Patients	select record	—	—	Patient Details
Patient Details	save	[version current]	update status and notes	Patients
Patient Details	cancel	—	—	Patients

From	Event	Guard	Action	To
any Driver function page	"Trips"	—	—	Driver Trips
any Driver function page	"Availability"	[vehicle registration approved]	—	Driver Availability
any Driver function page	"My Vehicle"	—	—	My Vehicle
Driver Trips	"View trip details"	—	—	Driver Trip Details
Driver Trip Details	close	—	—	Driver Trips
Driver Trip Details	"Report unavailability"	—	—	Confirm Unavailability
Confirm Unavailability	confirm	—	flag trip At-Risk	Driver Trip Details
Confirm Unavailability	cancel	—	—	Driver Trip Details
any Patient or Caregiver function page	"Trips"	—	—	Patient Trips
any Patient or Caregiver function page	"Schedules"	—	—	Patient Schedules
any Patient or Caregiver function page	"Caregivers"	[role = Patient]	—	Caregiver Connections
any Patient or Caregiver function page	"Patient"	[role = Caregiver]	—	Caregiver Connections
Patient Schedules	"Add Schedule" / "Edit Schedule"	—	—	Schedule Form
Schedule Form	save	[version current, no overlapping session]	store treatment schedule	Patient Schedules
Schedule Form	cancel	—	—	Patient Schedules
Schedule Form	"Delete Schedule"	[schedule owned]	—	Confirm Schedule Deletion
Confirm Schedule Deletion	confirm	—	delete schedule and linked replacements	Patient Schedules
Confirm Schedule Deletion	cancel	—	—	Schedule Form
Patient Trips	"View trip details"	—	—	Patient Trip Details
Patient Trip Details	close	—	—	Patient Trips
Patient Trip Details	"Cancel trip"	[trip upcoming]	—	Confirm Cancellation

From	Event	Guard	Action	To
Confirm Cancellation	confirm	—	release driver capacity	Patient Trip Details
Confirm Cancellation	cancel	—	—	Patient Trip Details
Patient Trip Details	"Rate driver"	[trip completed AND role = Patient]	—	Rate Driver
Rate Driver	submit	—	record rating	Patient Trip Details
Rate Driver	cancel	—	—	Patient Trip Details
Patient Trip Details	"Preferred driver"	[role = Patient]	—	Preferred Driver
Preferred Driver	save	—	set preferred driver	Patient Trip Details
Preferred Driver	remove	—	clear preferred driver	Patient Trip Details
Preferred Driver	cancel	—	—	Patient Trip Details
Patient Trips	"Add reminder"	—	—	Add Reminder
Add Reminder	confirm	—	send event to Calendar Service	Patient Trips
Add Reminder	cancel	—	—	Patient Trips
any function page (all role maps)	sign out	—	clear session	Login